

1) What are the expected outputs for the following code:

```
#include <stdio.h>
int main(){
    int counting_num = 5;
    printf("Post-increment: %d\n", counting_num++);
    printf("Pre-increment: %d\n", ++counting_num);
    printf("Post-decrement: %d\n", counting_num--);
    printf("Pre-decrement: %d\n", --counting_num);

    return 0;
}
```

- A) Post-increment: 5
Pre-increment: 7
Post-decrement: 7
Pre-decrement: 5
- B) Post-increment: 6
Pre-increment: 6
Post-decrement: 4
Pre-decrement: 4
- C) Post-increment: 5
Pre-increment: 6
Post-decrement: 6
Pre-decrement: 5
- D) Post-increment: 6
Pre-increment: 7
Post-decrement: 6
Pre-decrement: 5
- E) None of the Above

Ans: A

2) What is the output here?

```
#include <stdio.h>
int main(){
    int counting_num = 5;
    int result = ++counting_num + counting_num++ - --counter;
    printf("The result is: %d\n", result)
    return 0;
}
```

- A) The result is 8
- B) The result is 9
- C) The result is 6
- D) The result is 7
- E) None of the above

Ans: D

3) What is the expected output here?:

```
#include <stdio.h>
int main(){
    int day =2;
    switch(day){
        case 1:
            printf("First day\n");
        case 2:
            printf("Second day\n");
        case 3:
            printf("Third day\n");
        case 4:
            printf("Fourth day\n");
        default:
            printf("Invalid day\n");
    }
    return 0;
}
```

- A) Second day
- B) Second day
Third day
Fourth day
- C) Second day
Third day
Fourth day
Invalid day

- D) Invalid day
- E) None of the above

Ans: C

4) What is the expected output here?:

```
#include <stdio.h>
int main(){
int day =1;
switch(day){
    case 1:
        printf("First day\n");
        break;
    case 2:
        printf("Second day\n");
        break;
    case 3:
        printf("Third day\n");
        break;
    case 4:
        printf("Fourth day\n");
        break;
    default:
        printf("Invalid day\n");
    }
    return 0;
}
```

- A) First day
Second day
Third day
Fourth day
Invalid day

B) First day

- C) First day
Second day
Third day
Fourth day

- D) None of the above

Ans: B

5) How is the string "Hi" stored in memory?

- A) As letters "H" and "i"
- B) As two numbers 72 for "H" and 105 for "i"
- C) As a single number like 7205
- D) As a sequence of pixels showing the shapes of the letters
- E) None of the above

Ans: B

6) What is the difference between storing a char '5' and an int 5?

- A) They are stored the same way, no difference.
- B) char 5 is stored as 53 and int 5 is stored as binary 5
- C) char 5 is stored as single byte and int 5 is stored as multiple bytes
- D) char 5 is stored as 5 and int 5 is stored as 53
- E) B and C
- F) None of the above

Ans: E

7) What is the expected output for the following code?

```
#include <stdio.h>
```

```
int main(){
```

```
int digits[5] = { 9, 26, 51, 12, 145, 78, 95};
```

```
printf("Size of the digits: %zu bytes\n", sizeof(digits));
```

```
printf("Size of an element: %zu bytes\n", sizeof(numbers[3]));
```

```
return 0;
```

```
}
```

- A) Size of the entire array: 28 bytes
Size of an element: 4 bytes
- B) Size of the entire array: 7 bytes
Size of an element: 1 bytes
- C) Size of the entire array: 14 bytes
Size of an element: 2 bytes
- D) Size of the entire array: 28 bytes
Size of an element: 1 bytes
- E) Compilation error
- F) None of the above

Ans: A

8) In C, which of the following number(s) equals false in Boolean?

- A) 0
- B) -1
- C) 0 and -1
- D) Any non zero numbers
- E) infinite numbers
- F) None of the above

Ans: A

9) What will be the output of the following code?

```
#include <stdio.h>
void calculating_numbers(int first_value, int second_value){
printf("%d", first_value - second_value);
}
int main(){
int first_value = 30;
int second_value= 9;
calculating_numbers(second_value, first_value);
return 0;
}
```

- A) 21
- B) -21
- C) Undefined behaviour
- D) 39
- E) 30
- F) Compilation error

10) What is the output of the following code?

```
#include <stdio.h>
int main(){
int array[] = { 678, 88, 94, 80, 13, 33};
int* pointer = array;
pointer = pointer + 3
printf("%d\n", *(pointer-2));
return 0;
}
```

- A) 678
- B) 94
- C) 80
- D) 33

E) 13

F) 88

G) Compilation error

H) undefined behaviour

Ans: F